Knowledge-Grounded Reasoning and Self-Improvement for Autonomous AI Agents

Overview Autonomous AI agents must reason about unfamiliar situations, retrieve relevant knowledge, use external tools, and update decisions as evidence changes. Although large language models can generate sophisticated solution traces, their intermediate steps may rely on invented facts, ignore critical conditions, or use valid scientific rules in inappropriate settings. An agent can even reach a correct final answer through invalid reasoning. Because outcome-based learning primarily rewards the final result, it offers little guidance about which steps are reliable or how agents should recover from a failure.

We propose knowledge-grounded, self-improving reasoning agents that explicitly connect intermediate inferences to observations, structured relations, and verifiable rules. Our central hypothesis is that external knowledge representations and fine-grained process feedback, combined with tool-based checking, can improve transfer to unfamiliar tasks beyond outcome supervision alone. Three interrelated challenges motivate the project: (1) organizing heterogeneous knowledge and applicability conditions for compositional reasoning; (2) generating scalable, trustworthy process-level learning signals; and (3) deciding when and how to verify and repair an agent's intermediate assumptions.

We will pursue three interrelated research threads:

- **Knowledge acquisition and structured grounding:** We will retrieve and extract entities, events, relations, scientific principles, and contextual constraints from diverse texts. Lightweight evidence structures will preserve source provenance and distinguish reported facts from inferred links. In particular, the system will represent the conditions under which a principle or computational procedure is applicable, enabling agents to identify missing prerequisites before drawing conclusions.
- **Synthetic reasoning trajectories and process-guided learning:** We will synthesize multihop questions and reasoning sequences from structured knowledge, then generate controlled erroneous variants involving fabrication, context mismatch, invalid logic, and rule misuse. Fine-grained process reward models will classify errors at intermediate steps and provide feedback for candidate-step selection, iterative distillation, and policy improvement. We will study how reliably synthetic supervision transfers beyond the relations and domains used to construct training problems.
- **Verification-driven autonomous inquiry and revision:** Agents will formulate candidate conclusions or hypotheses, retrieve missing facts, select symbolic or executable checks, and diagnose failures. Rather than restarting after every failed attempt, an agent will update the affected assumption and propagate corrections through subsequent reasoning. Scientific hypothesis discovery provides an especially demanding testbed: the agent must retrieve unfamiliar literature, propose a plausible novelty, and test claims through executable or structured validation.

We will construct controlled task families in which familiar rules must be recombined under unfamiliar constraints and in which a previously useful scientific assumption is no longer valid. These cases will test whether improvements reflect genuine compositional reasoning or memorization of synthetic examples. We will separately measure the accuracy of the agent's stated assumptions, its choice of verification tool, and its ability to revise a partially correct trajectory while preserving valid reasoning steps. We will also study how the value of process feedback changes with limited data, noisy verification tools, and increasing task complexity.

Keywords: Autonomous agents, Knowledge grounding, Process reward models, Verifiable reasoning.

Intellectual Merit The research will advance compositional generalization, synthetic process supervision, and tool-aware reasoning under explicit constraints. We will measure intermediate-step validity, held-out knowledge transfer, successful error repair, final-task performance, and verification cost, comparing against retrieval-only agents, outcome-based training, conventional reward models, and unstructured tool use.

Naval Relevance and Broader Impact These foundational methods may improve the reliability of future Naval autonomous information-analysis and planning systems that must operate on incomplete knowledge and explain why an action is warranted. We will use public knowledge-intensive tasks and reproducible agent environments to support broader research reuse.